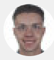
Corso: Data Science and Database Technology
Quiz: Exam 2025-01-20

PORTALE ESAMI

	[redacted]
Iniziato	20 gennaio 2025, 11:10
Stato	Completato
Terminato	20 gennaio 2025, 12:45
Tempo impiegato	1 ora 35 min.
Valutazione	**28,61** su un massimo di 32,00 (**89**%)
Riepilogo del tentativo	1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Domanda 1
Risposta corretta
Punteggio ottenuto 1,00 su 1,00

1 point (-15% penalty for each wrong answer)

Notation:

RN(V): read of object V by transaction N

WN(V): write of object V by transaction N

Text

The following schedule S of 3 transactions is given:

R3(z), R2(y), R3(x), R3(z), W2(z), R3(y), W1(z), W3(x), W1(x), R2(y)

Check whether S is conflict serializable and, if so, select the equivalent serial schedule.

- a. R2(y), W2(z), R2(y), R3(z), R3(x), R3(z), R3(y), W3(x), W1(z), W1(x)
- b. S is not conflict serializable
- c. R3(z), R3(x), R3(z), R3(y), W3(x), W1(z), W1(x), R2(y), W2(z), R2(y)
- d. W1(z), W1(x), R3(z), R3(x), R3(z), R3(y), W3(x), R2(y), W2(z), R2(y)
- e. W1(z), W1(x), R2(y), W2(z), R2(y), R3(z), R3(x), R3(z), R3(y), W3(x)
- f. R2(y), W2(z), R2(y), W1(z), W1(x), R3(z), R3(x), R3(z), R3(y), W3(x)
- g. R3(z), R3(x), R3(z), R3(y), W3(x), R2(y), W2(z), R2(y), W1(z), W1(x) ✔

Risposta corretta.

La risposta corretta è:
R3(z), R3(x), R3(z), R3(y), W3(x), R2(y), W2(z), R2(y), W1(z), W1(x)

Domanda 2
Risposta non data
Non valutata

This question is not a part of the exam

You can use the text area below to write any note or draft (e.g. intermediate steps of an exercise).

Any text written below will not be considered toward the correction of the exam.

Domanda 3
Risposta corretta
Punteggio ottenuto 1,00 su 1,00

1 point (-15% penalty for each wrong answer)

Notation

Tn: Id of the transaction n
B(Tn): Begin of the transaction Tn
CK(Ta,Tb,...): checkpoint with unfinished Ta, Tb, ... transactions
C(Tn): commit of transaction Tn
A(Tn): abort (rollback) of transaction Tn
Un(x): update executed by transaction Tn on object x
In(x): insert executed by transaction Tn on object x
Dn(x): delete executed by transaction Tn on object x

The following sequence of operations within a log file is given:

B(T0), U0(O1), C(T0), CK(), CK(), B(T2), D2(O3), U2(O3), I2(O1), CK(T2), B(T3), I3(O0), C(T3), B(T1), A(T1), B(T4), U4(O0), FAILURE

What are the final UNDO and REDO sets for the warm restart?

a. REDO={T3}, UNDO={}

b. No answer is correct

c. REDO={}, UNDO={T1, T2, T4}

d. REDO={T3}, UNDO={T1}

e. REDO={}, UNDO={T1}

f. REDO={}, UNDO={T2, T4}

g. REDO={T3}, UNDO={T1, T2, T4} ✔

Risposta corretta.

La risposta corretta è:
REDO={T3}, UNDO={T1, T2, T4}

Domanda 4
Risposta corretta
Punteggio ottenuto 1,00 su 1,00

Conceptual schema (1 point, -15% penalty for each wrong answer)

We want to analyze information about food drives organized during the year on the Italian territory by a charity. These events aim at collecting and subsequent redistribution of food items to charitable organizations in the territory.

Food drives are held several times a year, specifying the types of products to be collected, such as baby food and/or canned tuna and meat. To support the food drive, customers can purchase products at various stores. The purchased products are then handed over to volunteers outside the store. Customers who have purchased products can then respond to a short interview (anonymously and not compulsorily) for the purpose of collecting certain statistical information such as the customer's age, how the customer learned about the initiative (e.g., through social platforms and/or through the press), whether the customer is a member of the association, and the customer's occupation (e.g., whether the customer is a student, employee, or retiree).

In addition to the support received from people who buy the products, each food drive also benefits from the support of some organizations that donate funds to the food drive, such as banking foundations, companies, etc.

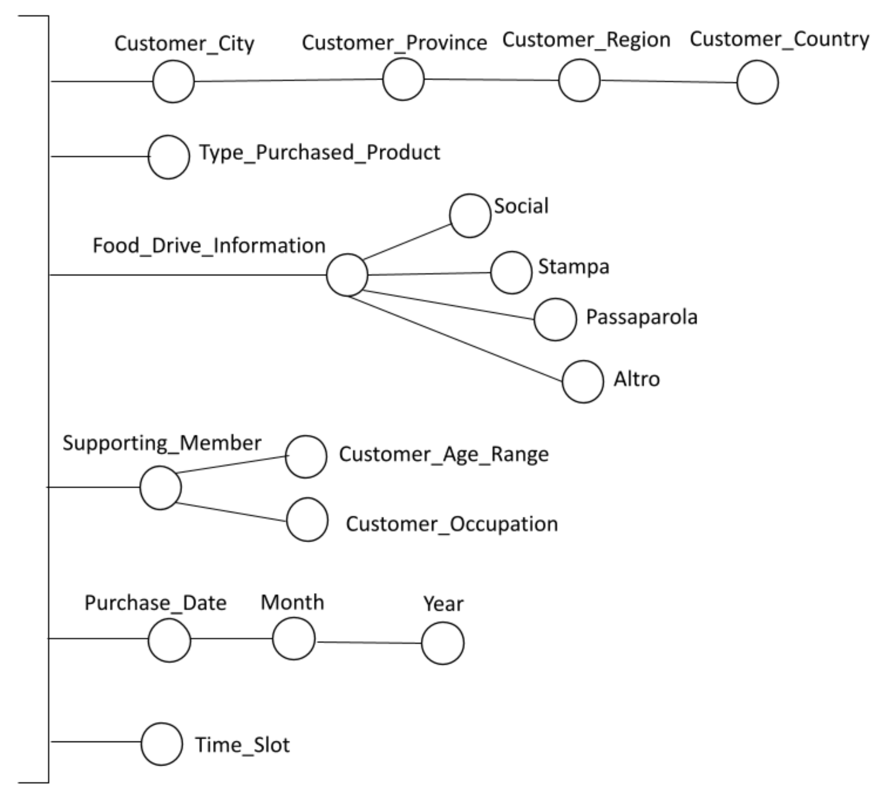
For each food drive, the month in which it is carried out is known, together with the indication of the corresponding 2-month period, 3-month period, 6-month period and year; the duration of the food drive in days is also known.

We want to analyze the average amount spent per customer and the average quantity purchased per customer based on the following information:

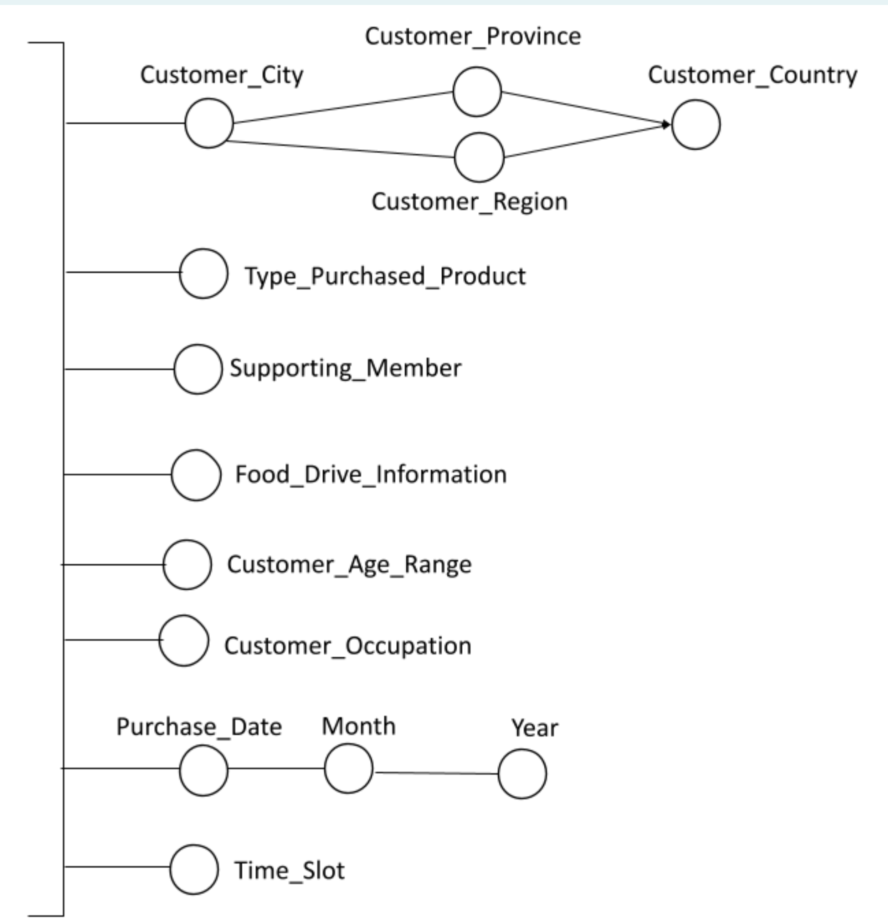
- food drives carried out. Each food drive is characterized by the type of products requested (one or more values among baby food, canned vegetables and legumes, canned tuna and meat) and by the list of organizations that supported the food drive (such as banking foundations, companies, etc.). For each food drive, the month in which it is carried out is known, together with the indication of the corresponding 2-month period, 3-month period, 6-month period, and year; the duration of the food drive in days is also known.
- stores where customers can buy products to support the food drive. For each store, the geographical location is known in terms of city district, city, province and region. The size of the store (a value between small, medium, large) and the supermarket chain to which it belongs are also known.
- type of product purchased by the customer (a value between baby food, canned vegetables and legumes, canned tuna and meat)
- indication whether the customer is a supporting member of the charity or not (a value between supporter and non-supporter)
- city, province, region and country of the customer who made the purchase
- indication of how the customer learned about the food drive (one or more values among social platforms, press, word of mouth, other)
- customer age range (a value between under 21, between 21 and 40, between 41 and 60, over 60) and customer occupation (a value between student, employee, retiree)
- date, month, year in which the customer made the purchase and the time slot (a value between morning, early afternoon, evening)

Select, from the dimensions proposed below, those that meet the requirements described in the problem specifications.

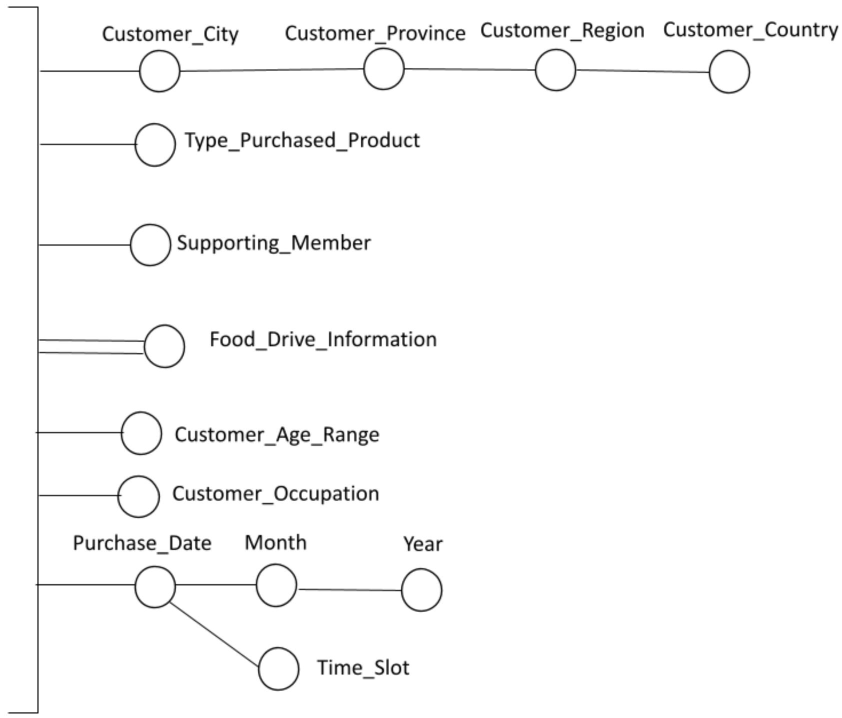
a.



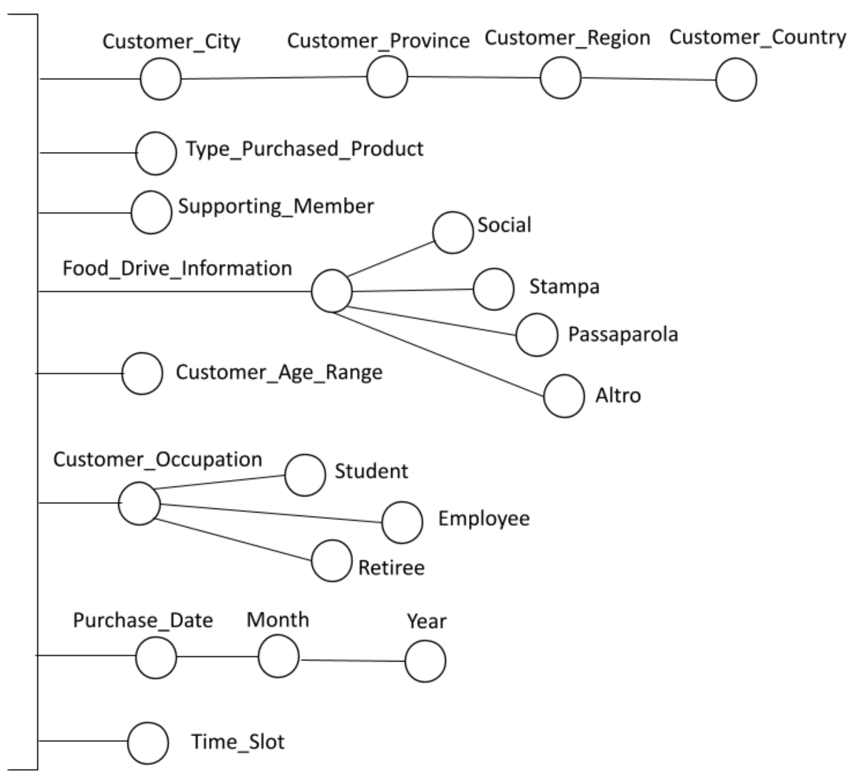
b.



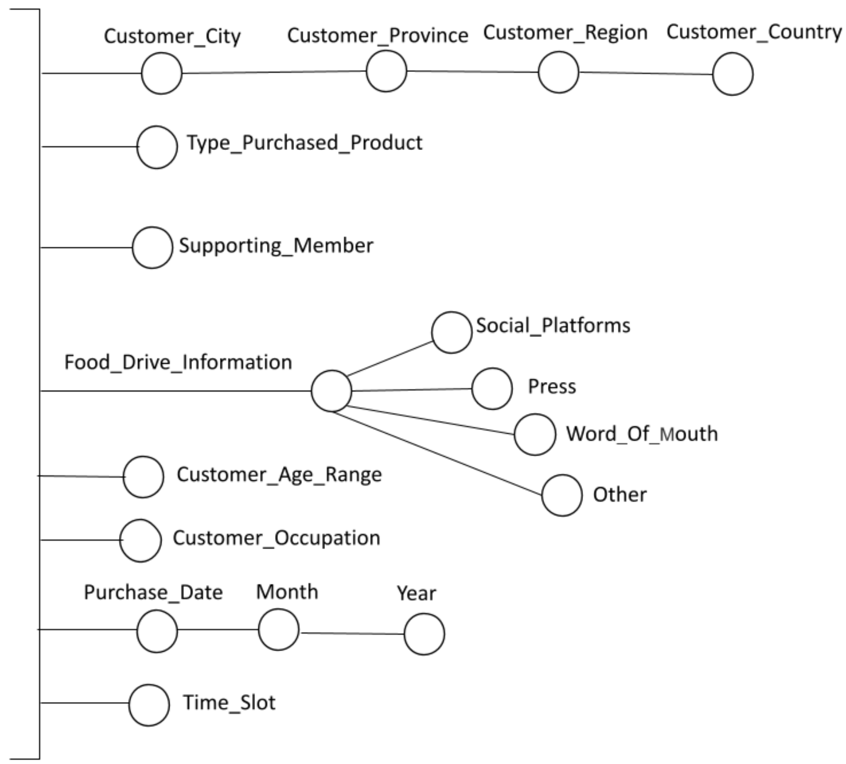
c.


Customer_City
Customer_Province
Customer_Region
Customer_Country
Type_Purchased_Product
Supporting_Member
Food_Drive_Information
Customer_Age_Range
Customer_Occupation
Purchase_Date
Month
Year
Time_Slot

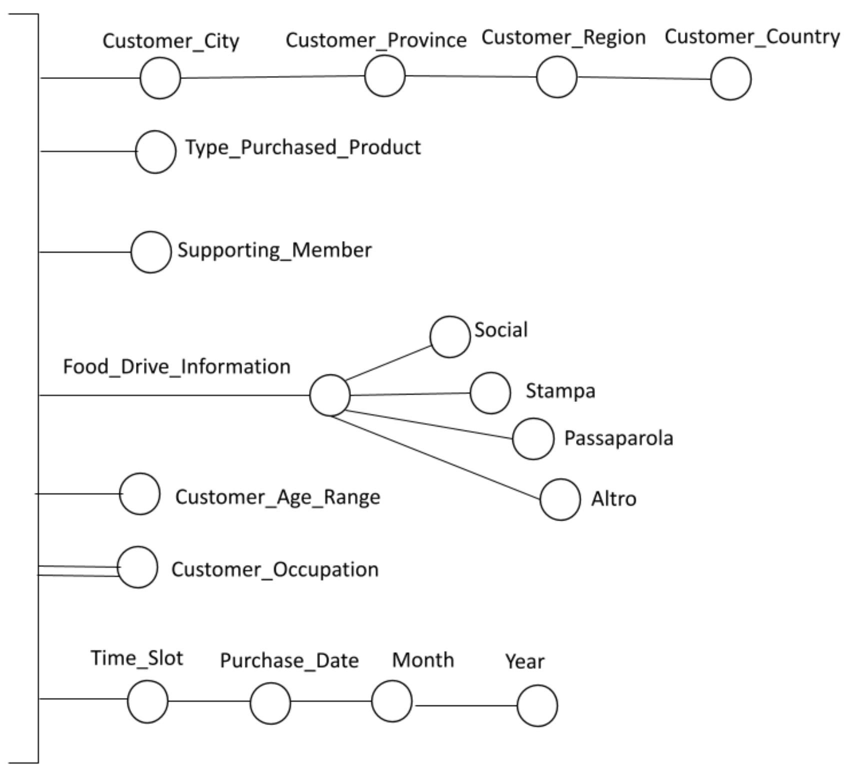
d.


Customer_City
Customer_Province
Customer_Region
Customer_Country
Type_Purchased_Product
Supporting_Member
Social
Food_Drive_Information
Stampa
Passaparola
Customer_Age_Range
Altro
Customer_Occupation
Student
Employee
Retiree
Purchase_Date
Month
Year
Time_Slot

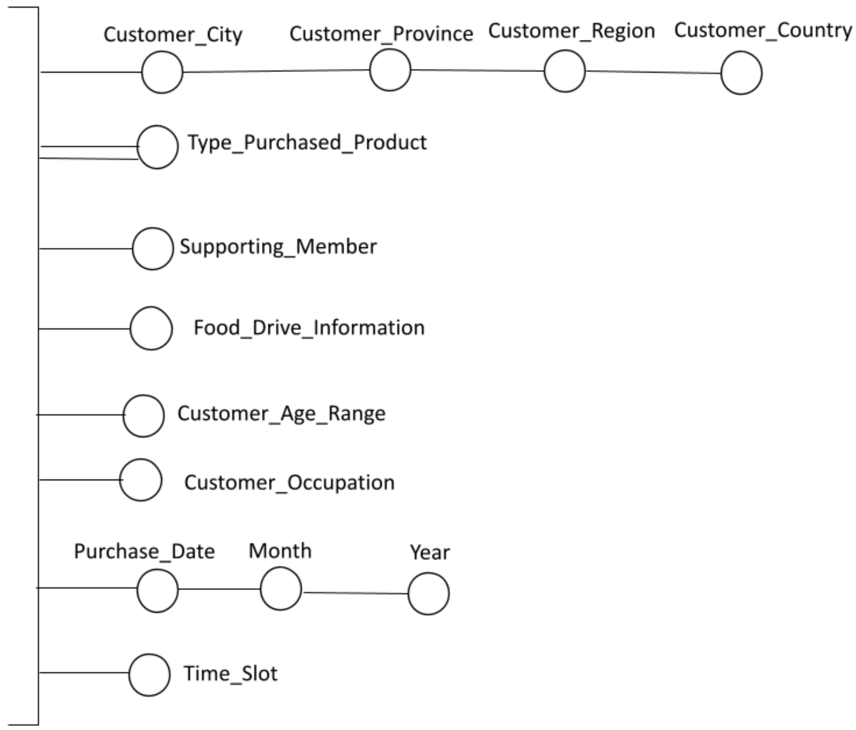
e.


Customer_City
Customer_Province
Customer_Region
Customer_Country
Type_Purchased_Product
Supporting_Member
Social_Platforms
Food_Drive_Information
Press
Word_Of_Mouth
Customer_Age_Range
Other
Customer_Occupation
Purchase_Date
Month
Year
Time_Slot

f.

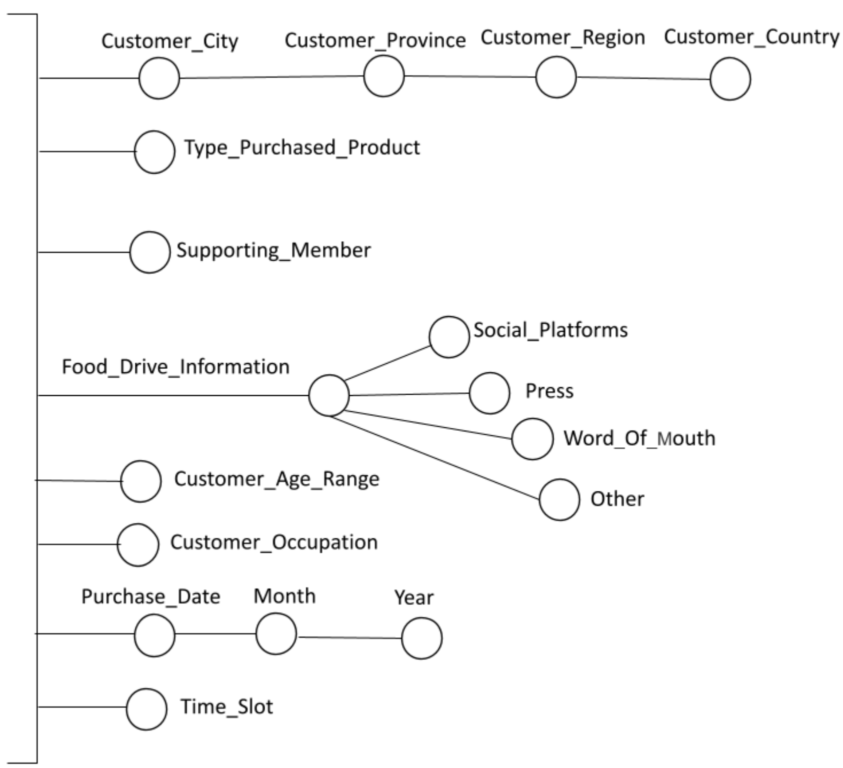

Customer_City
Customer_Province
Customer_Region
Customer_Country
Type_Purchased_Product
Supporting_Member
Social
Food_Drive_Information
Stampa
Passaparola
Customer_Age_Range
Altro
Customer_Occupation
Time_Slot
Purchase_Date
Month
Year

g.



Risposta corretta.

La risposta corretta è:



Extended SQL (4 points)

The following data warehouse stores information on car rentals made by an international car rental company (the cars are characterized by model, brand, and presence of accessories). The presence of accessories (e.g., navigation system) is modeled as a multiple edge, subsequently represented by the table ACCESSORIES. The

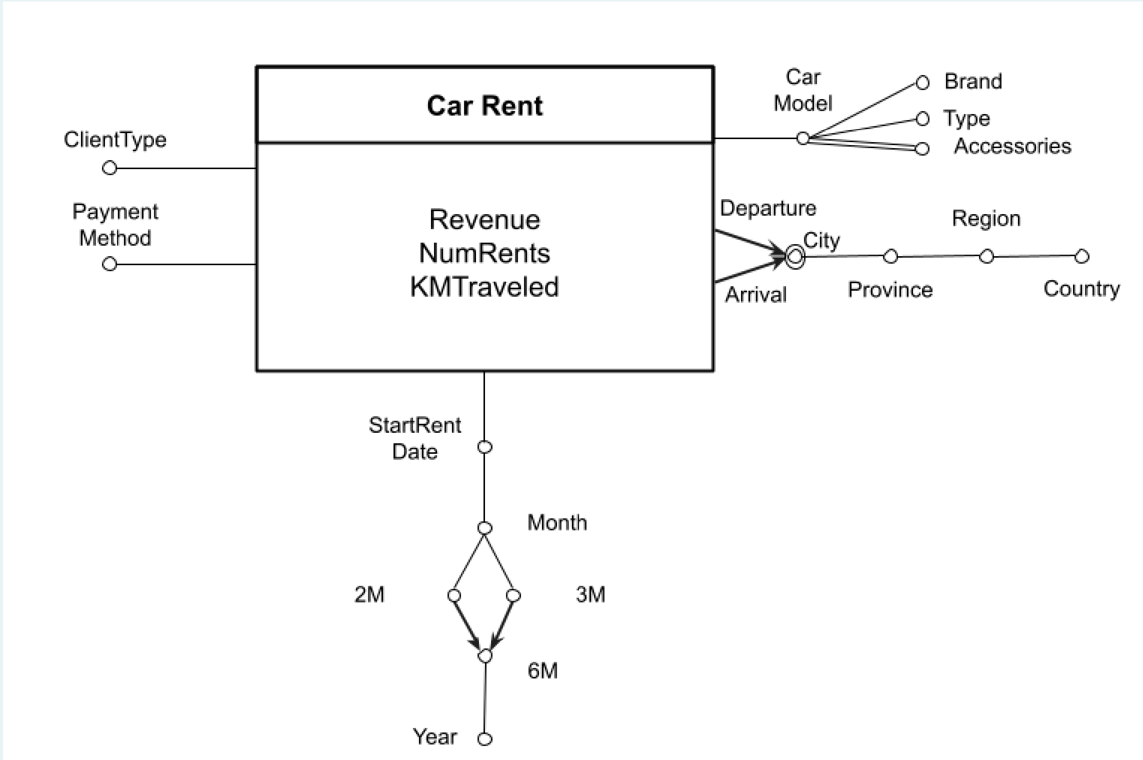
Domanda **5**

Completo

Punteggio ottenuto 4,00 su 4,00

rental is also described by a payment method, the type of client who made the rental, the start date of the rental, the city of departure, and the city of arrival of the rental.

The data warehouse is characterized by the following conceptual schema and corresponding logical schema. Each event is characterized by three measures: collection, number of rentals, and miles traveled.



CAR-MODEL (IDModel, Model, Type, Brand)
ACCESSORIES (IDModel, Accessory)
TIME (IDTime, StartDate, Month, 2M, 3M, 6M, Year)
JUNK-RENT (IDJR, ClientType, PaymentMethod)
CITY (IDCity, City, Province, Region, Country)
CAR-RENT (IDModel, IDTime, IDJR, IDDepartureCity, IDArrivalCity, Revenue, NumRents, KMTraveled)

Considering only rentals of BMW cars (Brand='BMW') carried out in the period 2021-2024 compute for each bimonth (2-Month attribute) and departure city:

- the average revenue per rental and the average number of kilometers traveled per rental,
- the cumulative value of the revenue as bimonths pass separately by year,
- the percentage of the number of kilometers traveled compared to the total number of kilometers traveled by departure country.

Perform the analysis separately by client type.

```
SELECT
    JR.ClientType, T.2-Months, C.City,
    SUM(CR.Revenue)/SUM(CR.NumRents), SUM(CR.KMTraveled)/SUM(CR.NumRents),
    SUM(SUM(CR.Revenue)) OVER(PARTITION BY JR.ClientType, C.City, T.Year ORDER BY T.2-Months ROWS
UNBOUNDED PRECEDING),
    100 * SUM(CR.KMTraveled)/SUM(SUM(CR.KMTraveled)) OVER(PARTITION BY JR.ClientType, C.Country, T.2-
Months)
FROM
   CAR-RENT CR, JUNK-RENT JR, CAR-MODEL CM, TIME T, CITY C
WHERE
   CR.IDJR=JR.IDJR AND
```

```
    CR.IDModel=CM.IDModel AND
    CR.IDTime=T.IDTime AND
    CR.IDDepartureCity=C.IDCity AND
    CM.Brand='BMW' AND
    (T.Year>='2021' AND T.Year<='2024')
GROUP BY
    JR.ClientType, T.2-Months, C.City, T.Year, C.Country
```

Commento:

Domanda 6
Risposta corretta
Punteggio ottenuto 1,00 su 1,00

Measures (1 point, -15% penalty for each wrong answer)

We want to analyze information about food drives organized during the year on the Italian territory by a charity. These events aim at collecting and subsequent redistribution of food items to charitable organizations in the territory.

Food drives are held several times a year, specifying the types of products to be collected, such as baby food and/or canned tuna and meat. To support the food drive, customers can purchase products at various stores. The purchased products are then handed over to volunteers outside the store. Customers who have purchased products can then respond to a short interview (anonymously and not compulsorily) for the purpose of collecting certain statistical information such as the customer's age, how the customer learned about the initiative (e.g., through social platforms and/or through the press), whether the customer is a member of the association, and the customer's occupation (e.g., whether the customer is a student, employee, or retiree).

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For each food drive, the month in which it is carried out is known, together with the indication of the corresponding 2-month period, 3-month period, 6-month period and year; the duration of the food drive in days is also known.

We want to analyze the average amount spent per customer and the average quantity purchased per customer based on the following information:

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corresponding 2-month period, 3-month period, 6-month period, and year; the duration of the food drive in days is also known.
- stores where customers can buy products to support the food drive. For each store, the geographical location is known in terms of city district, city, province and region. The size of the store (a value between small, medium, large) and the supermarket chain to which it belongs are also known.
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- city, province, region and country of the customer who made the purchase
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- customer age range (a value between under 21, between 21 and 40, between 41 and 60, over 60) and customer occupation (a value between student, employee, retiree)
- date, month, year in which the customer made the purchase and the time slot (a value between morning, early afternoon, evening)

Select from the list below all and only the attributes required to correctly model the requests in the specifications for the fact table (multiple answers can be correct, since multiple measurements can be required)

- a. Average quantity purchased, Total number of customers, Total amount spent
- b. Total quantity purchased, Total number of food drives, Total amount spent
- c. Total quantity purchased, Total number of food drives, Minimum amount spent
- d. Total quantity purchased, Total number of food drives, Maximum amount spent
- e. Total quantity purchased, Total number of customers, Maximum amount spent
- f. Average quantity purchased, Total number of food drives, Maximum amount spent
- g. Average quantity purchased, Total number of customers, Maximum amount spent
- h. Total quantity purchased, Total number of customers, Total amount spent ✔

Risposta corretta.

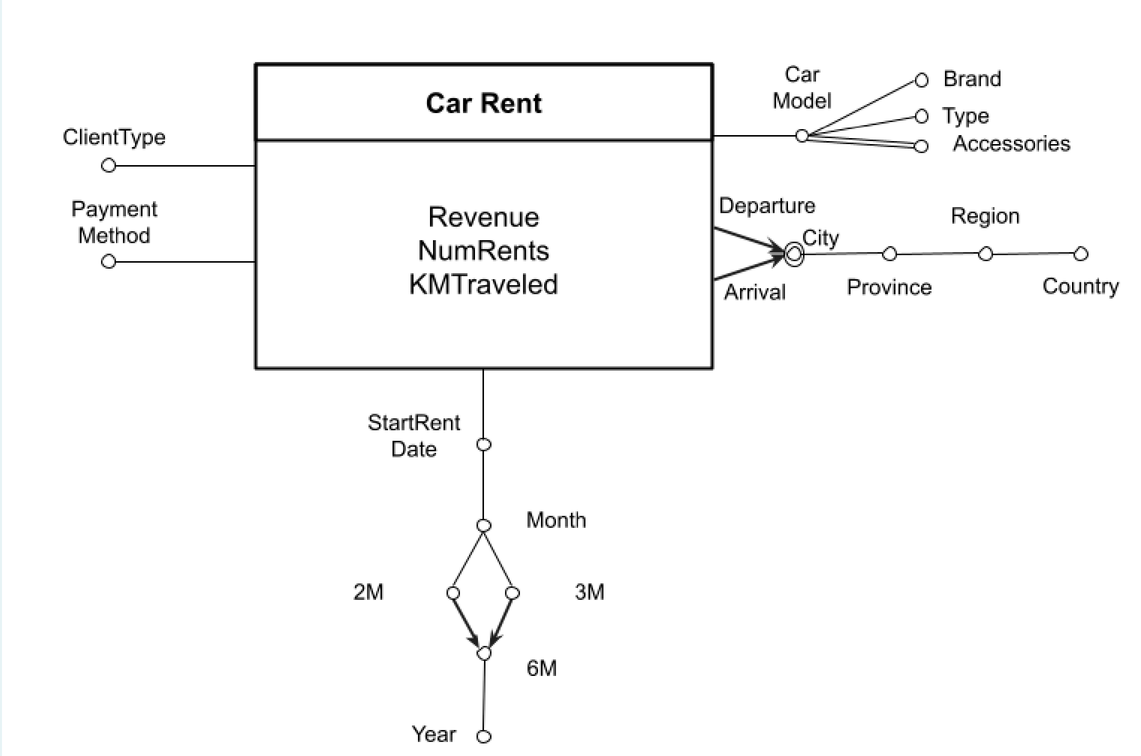
La risposta corretta è:
Total quantity purchased, Total number of customers, Total amount spent

Domanda 7
Completo
Punteggio ottenuto 3,00 su 3,00

Extended SQL (3 points)

The following data warehouse stores information on car rentals made by an international car rental company (the cars are characterized by model, brand, and presence of accessories). The presence of accessories (e.g., navigation system) is modeled as a multiple edge, subsequently represented by the table ACCESSORIES. The rental is also described by a payment method, the type of client who made the rental, the start date of the rental, the city of departure, and the city of arrival of the rental.

The data warehouse is characterized by the following conceptual schema and corresponding logical schema. Each event is characterized by three measures: collection, number of rentals, and miles traveled.



CAR-MODEL (IDModel, Model, Type, Brand)
ACCESSORIES (IDModel, Accessory)
TIME (IDTime, StartDate, Month, 2M, 3M, 6M, Year)
JUNK-RENT (IDJR, ClientType, PaymentMethod)
CITY (IDCity, City, Province, Region, Country)
CAR-RENT (IDModel, IDTime, IDJR, IDDepartureCity, IDArrivalCity, Revenue, NumRents, KMTraveled)

Considering rentals of cars with navigator accessory (Accessory='Navigator'), for each semester and car model show

- the average revenue per number of rentals,
- the percentage of total revenue with respect to the total revenue by car brand and year.

Assign to each record:

- a rank based on the number of rentals made (rank 1 for the highest number of rentals) separately by car brand
- a rank based on the number of kilometers traveled (position 1 for the lowest number of kilometers traveled) separately by year.

```
SELECT
    T.6-Months, CM.Model,
    SUM(CR.Revenue)/SUM(CR.NumRents),
    100 * SUM(CR.Revenue)/SUM(SUM(CR.Revenue)) OVER(PARTITION BY CR.Brand, T.Year) ,
    RANK() OVER(PARTITION BY CM.Brand ORDER BY SUM(CR.NumRents) DESC),
    RANK() OVER(PARTITION BY T.Year ORDER BY SUM(CR.KMTraveled)
FROM
   CAR-RENT CR, ACCESSORIES A, TIME T, CAR-MODEL CM
WHERE
   CR.IDModel=A.IDModel AND
   CR.IDTime=T.IDTime AND
   CR.IDModel=CM.IDModel AND
```

A.Accessory='Navigator'

GROUP BY

T.6-Months, CM.Model, CR.Brand, T.Year

Commento:

Domanda 8

Risposta corretta

Punteggio ottenuto 5,00 su 5,00

Cardinalities (1.5 points, -15% penalty for each wrong answer)

The following tables are provided:

PROFESSOR(<u>PID</u>, FirstName, LastName, Department, Salary)

COURSE(<u>CID</u>, CourseName, CourseType, SubjectArea, Requirements)

PROFESSOR-TEACHES(<u>PID</u>, <u>CID</u>, <u>DateAssigned</u>)

STUDENT(<u>SID</u>, FirstName, LastName, BirthDate, BirthCountry)

ENROLLMENT(<u>SID</u>, <u>CID</u>, CreditsEarned, EnrollmentDate)

Assume the following cardinalities:

- card(PROFESSOR) = $5 \cdot 10^3$ tuples
  distinct values of Department = 10
- card(COURSE) = $2 \cdot 10^2$ tuples
  distinct values of SubjectArea = 20
- card(PROFESSOR-TEACHES) = $4 \cdot 10^5$ tuples
  MIN(DateAssigned) = 1/1/2005, MAX(DateAssigned) = 31/12/2024
- card(STUDENT) = $5 \cdot 10^4$ tuples
  distinct values of BirthCountry = 100
  MIN(BirthDate) = 1/1/1990, MAX(BirthDate) = 31/12/2004
- card(ENROLLMENT) = $2 \cdot 10^5$ tuples

Furthermore, assume the following reduction factor for the having clauses:

HAVING AVERAGE(P.Salary) > 30000 = 1/3

Consider the following query:

SELECT PT.CID, COUNT(*)

FROM PROFESSOR P, COURSE C, PROFESSOR-TEACHES PT

WHERE P.PID = PT.PID

AND C.CID = PT.CID

AND C.SubjectArea = 'Computer Engineering'
AND PT.DateAssigned >= 1/1/2024

AND P.Department = 'DAUIN'

AND PT.CID IN (

SELECT E.CID

FROM ENROLLMENT E, STUDENT S

WHERE E.SID = S.SID

AND S.BirthCountry = 'Japan'

AND S.BirthDate <= 1/1/2004

)

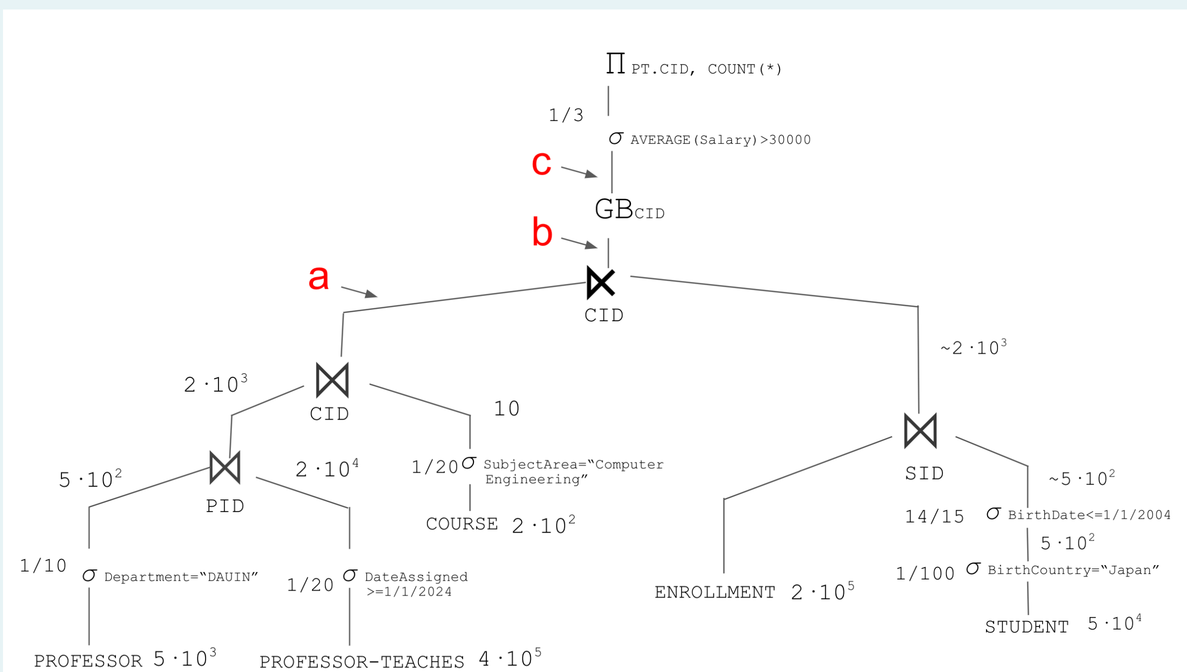
GROUP BY PT.CID

HAVING AVERAGE(P.Salary) > 30000;

Cardinality

(1.5 pt, -15% penalty for each wrong answer)

The figure below represents the query tree for the query above.



Select the correct answer for the cardinality of **(a)**:

- 10^2 ✔
- $2 \cdot 10^2$
- $2 \cdot 10^3$
- 10^3

Punteggio ottenuto 5,00 su 5,00

La risposta corretta è: 10^2

Select the correct answer for the cardinality of **(b)**:

- $\sim 10^3$
- $\sim 10^2$ ✔
- $\sim 2 \cdot 10$
- $\sim 2 \cdot 10^3$

Punteggio ottenuto 5,00 su 5,00

La risposta corretta è: ~10^2

Select the correct answer for the cardinality of **(c)**:

- ○ ~$2 \cdot 10^2$
- ○ ~$2 \cdot 10$
- ○ ~10^2
- ◉ ~10 ✔

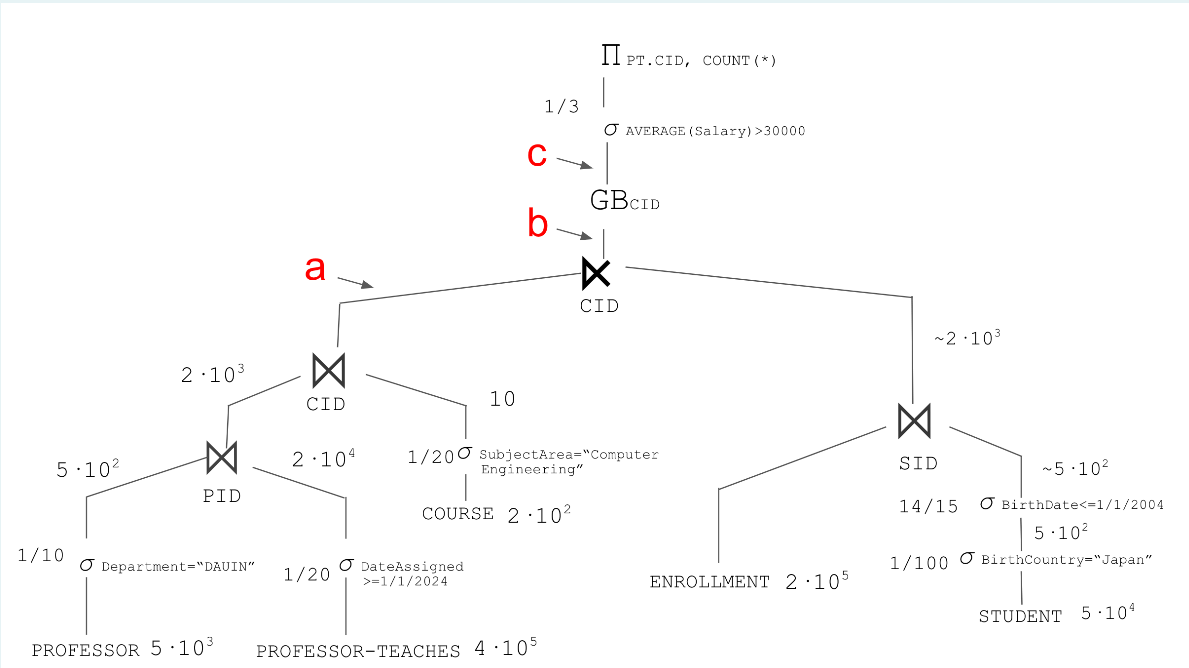
Punteggio ottenuto 5,00 su 5,00

La risposta corretta è: ~10

Indexes

(1.5 pt, -15% penalty for each wrong answer)

The figure below represents the query tree for the query above.



Select, for each table, accessory physical structures to improve query performance (if possible) from the options below.

Table PROFESSOR

- ◉ CREATE INDEX IndexA ON PROFESSOR(Department) - HASH ✔
- ○ CREATE INDEX IndexB ON PROFESSOR(Department) - B+-Tree
- ○ An index on this attribute would not improve query performance

Punteggio ottenuto 3,00 su 3,00

La risposta corretta è: CREATE INDEX IndexA ON PROFESSOR(Department) - HASH

Table PROFESSOR-TEACHES

- ○ CREATE INDEX IndexC ON PROFESSOR-TEACHES(AssignmentDate) - HASH
- ◉ CREATE INDEX IndexD ON PROFESSOR-TEACHES(AssignmentDate) - B+-Tree ✔
- ○ An index on this attribute would not improve query performance

Punteggio ottenuto 3,00 su 3,00

La risposta corretta è: CREATE INDEX IndexD ON PROFESSOR-TEACHES(AssignmentDate) - B+-Tree

Table COURSE

- ○ CREATE INDEX IndexE ON COURSE(TopicArea) - HASH
- ○ CREATE INDEX IndexF ON COURSE(TopicArea) - B+-Tree
- ◉ An index on this attribute would not improve query performance ✔

Punteggio ottenuto 3,00 su 3,00

La risposta corretta è: An index on this attribute would not improve query performance

Table STUDENT

- ○ CREATE INDEX IndexG ON STUDENT(BirthDate) - HASH
- ○ CREATE INDEX IndexH ON STUDENT(BirthDate) - B+-Tree
- ◉ An index on this attribute would not improve query performance ✔

Punteggio ottenuto 3,00 su 3,00

La risposta corretta è: An index on this attribute would not improve query performance

Table STUDENT

- ◉ CREATE INDEX IndexI ON STUDENT(BirthCountry) - HASH ✔
- ○ CREATE INDEX IndexJ ON STUDENT(BirthCountry) - B+-Tree
- ○ An index on this attribute would not improve query performance

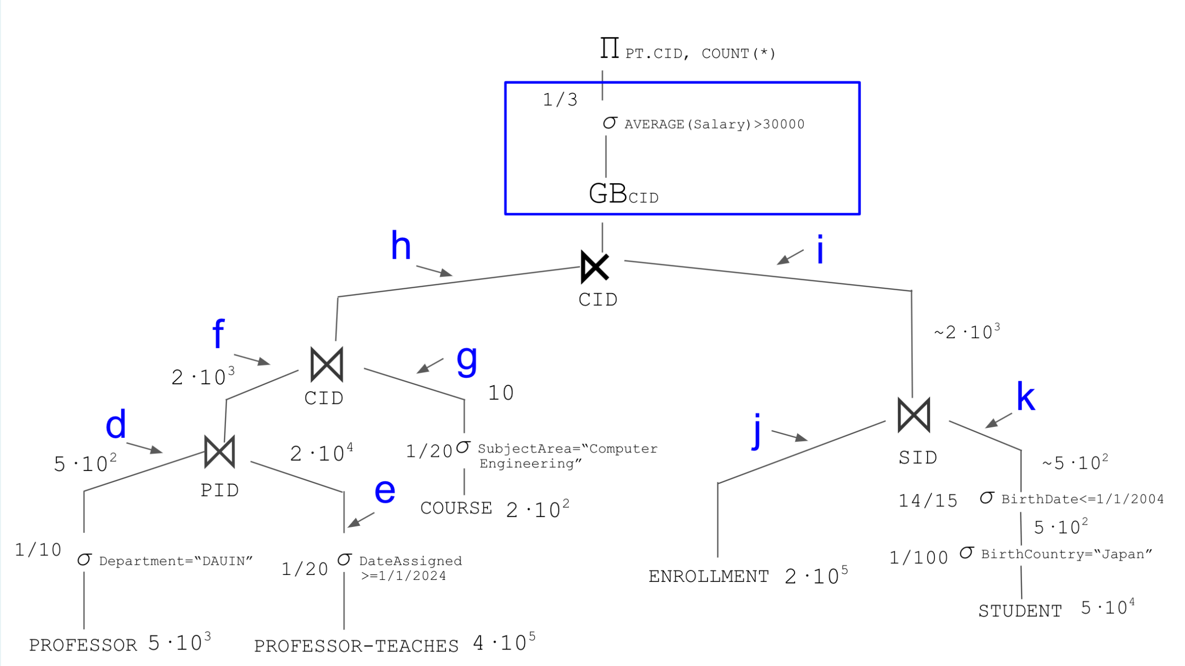
Punteggio ottenuto 3,00 su 3,00

La risposta corretta è: CREATE INDEX IndexI ON STUDENT(BirthCountry) - HASH

Anticipate Group By

(2 pt, -15% penalty for each wrong answer)

The figure below represents the query tree for the query above.



Analyze the anticipation of the **GROUP BY GROUP BY PI.CID HAVING AVERAGE(P.Salary) > 30000** represented in the box.
Select the solution that **allows for the maximum efficiency** in executing the query (if it exists)

- It is possible to anticipate it in branch d
- It is possible to anticipate it in branch e
- It is possible to anticipate it in branch f ✔
- It is possible to anticipate it in branch g
- It is possible to anticipate it in branch h
- It is possible to anticipate it in branch i
- It is possible to anticipate it in branch j
- It is possible to anticipate it in branch k
- It is not possible to anticipate the Group BY GROUP BY PI.CID HAVING AVERAGE(P.Salary) > 30000

Punteggio ottenuto 20,00 su 20,00

La risposta corretta è: It is possible to anticipate it in branch f

Domanda 9
Risposta corretta
Punteggio ottenuto 1,50 su 1,50

1.5 points (-15% penalty for each wrong answer)

A rule-based classifier is provided with the following rules to classify animals:

1. If Habitat = Water, then Class = Fish
2. If Habitat = Water and Number of legs > 6, then Class = Crustacean
3. If Number of legs >= 2, then Class = Mammalian
4. If Number of legs = 8, then Class = Arachnid
5. If Can fly = Yes, then Class = Bird
6. If Habitat = Water and Number of legs = 4, then Class = Amphibious

A new animal with these attributes is found:

- Habitat: Land
- Number of legs: 4
- Can fly: No

To which class will this animal be assigned?

- a. Mammalian ✔
- b. Bird
- c. Fish
- d. Amphibious
- e. Arachnid
- f. No answer is correct
- g. Crustacean

Risposta corretta.

La risposta corretta è:
Mammalian

Domanda **10**

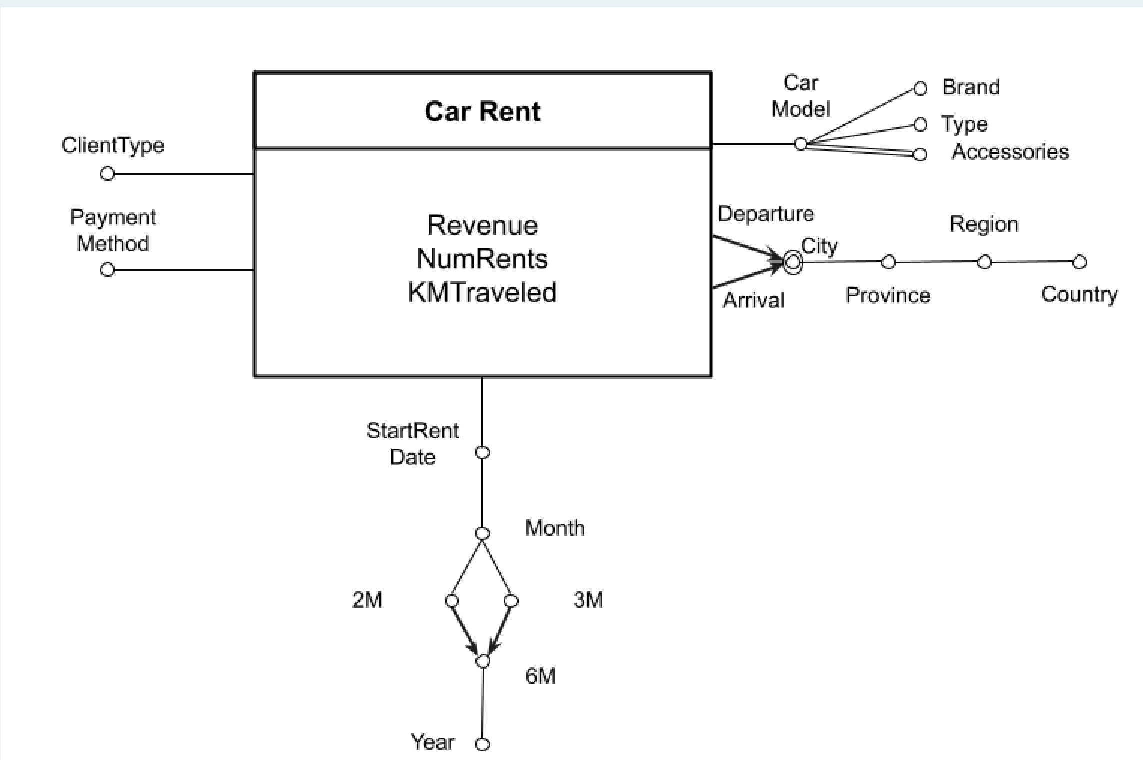
Completo

Punteggio ottenuto 4,00 su 4,00

Extended SQL (4 points)

The following data warehouse stores information on car rentals made by an international car rental company (the cars are characterized by model, brand, and presence of accessories). The presence of accessories (e.g., navigation system) is modeled as a multiple edge, subsequently represented by the table ACCESSORIES. The rental is also described by a payment method, the type of client who made the rental, the start date of the rental, the city of departure, and the city of arrival of the rental.

The data warehouse is characterized by the following conceptual schema and corresponding logical schema. Each event is characterized by three measures: collection, number of rentals, and miles traveled.



CAR-MODEL (<u>IDModel</u>, Model, Type, Brand)
ACCESSORIES (<u>IDModel</u>, <u>Accessory</u>)
TIME (<u>IDTime</u>, StartDate, Month, 2M, 3M, 6M, Year)
JUNK-RENT (<u>IDJR</u>, ClientType, PaymentMethod)
CITY (<u>IDCity</u>, City, Province, Region, Country)
CAR-RENT (<u>IDModel</u>, <u>IDTime</u>, <u>IDJR</u>, <u>IDDepartureCity</u>, <u>IDArrivalCity</u>, Revenue, NumRents, KMTraveled)

For each triplet (province of rental departure city, province of rental arrival city, quarter (3-Months)), show

- the average revenue per traveled km,
- the percentage of the total revenue to the total revenue by region of the departure city and region of the arrival city
- the average revenue per number of rentals separately by year

Perform the analysis separately by payment method

```
SELECT
    JR.PaymentMethod, C1.Province, C2.Province, T.3-Months,
    SUM(CR.Revenue)/SUM(CR.KMTraveled),
    100 * SUM(CR.Revenue)/SUM(SUM(CR.Revenue)) OVER(PARTITION BY C1.Region, C2.Region, T.3-Months,
JR.PaymentMethod),
     [SUM(SUM(CR.Revenue))/SUM(SUM(CR.NumRents))] OVER(PARTITION BY T.Year, C1.Province, C2.Province,
JR.PaymentMethod)
FROM
    CAR-RENT CR, CITY C1, CITY C2, TIME T, JUNK-RENT JR
WHERE
    CR.IDDepartureCity=C1.IDCIty AND
    CR.IDArrivalCity=C2.IDCity AND
    CR.IDTime=T.IDTime AND
    CR.IDJR=T.IDJR
GROUP BY
    JR.PaymentMethod, C1.Province, C2.Province, T.3-Months, C1.Region, C2.Region, T.Year
```

Commento:

Domanda **11**

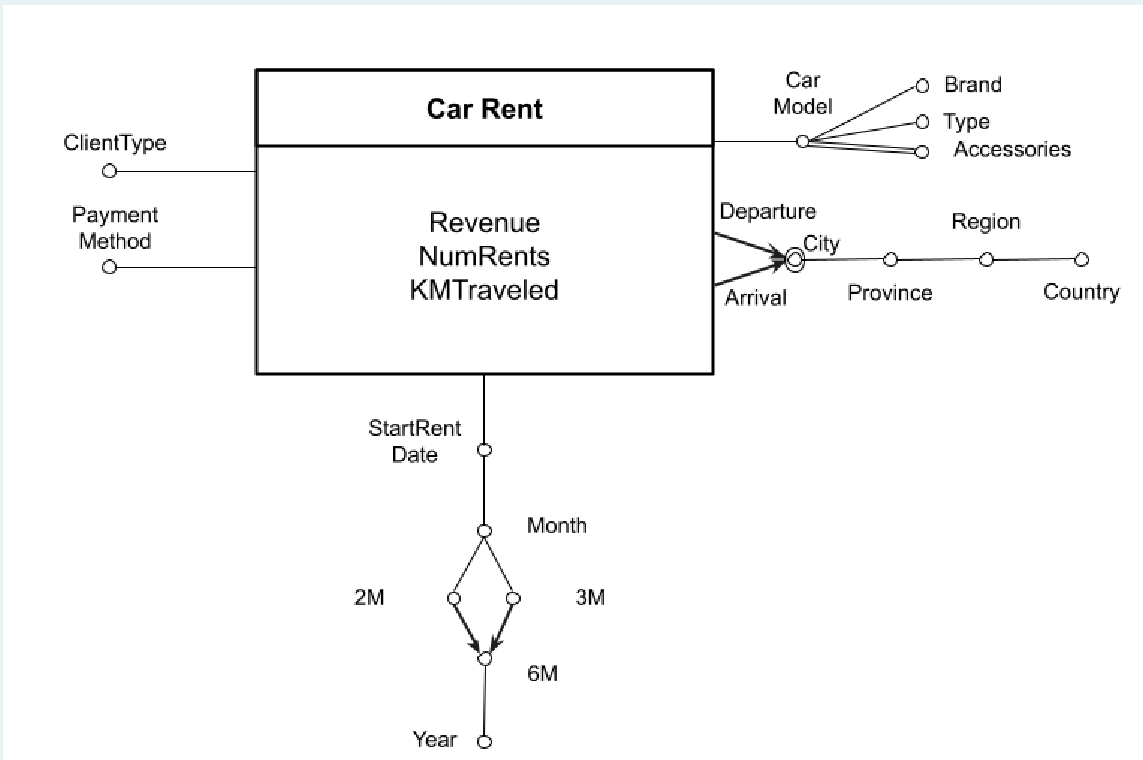
Completo

Punteggio ottenuto 4,75 su 5,00

Materialized view (5 points)

The following data warehouse stores information on car rentals made by an international car rental company (the cars are characterized by model, brand, and presence of accessories). The presence of accessories (e.g., navigation system) is modeled as a multiple edge, subsequently represented by the table ACCESSORIES. The rental is also described by a payment method, the type of client who made the rental, the start date of the rental, the city of departure, and the city of arrival of the rental.

The data warehouse is characterized by the following conceptual schema and corresponding logical schema. Each event is characterized by three measures: collection, number of rentals, and miles traveled.



CAR-MODEL (<u>IDModel</u>, Model, Type, Brand)
ACCESSORIES (<u>IDModel</u>, <u>Accessory</u>) TIME (<u>IDTime</u>, StartDate, Month, 2M, 3M, 6M, Year) JUNK-RENT (<u>IDJR</u>, ClientType, PaymentMethod) CITY (<u>IDCity</u>, City, Province, Region, Country) CAR-RENT (<u>IDModel</u>, <u>IDTime</u>, <u>IDJR</u>, <u>IDDepartureCity</u>, <u>IDArrivalCity</u>, Revenue, NumRents, KMTraveled)

Given the above logical scheme, consider the following queries of interest:

- Considering only 'Audi' brand car models and 'business' type clients, show the cumulative value of the number of rentals as the quarters pass (3-Months attribute), separately by year.
- For rentals departed from cities in the Piedmont region, separately by payment mode (attribute PaymentMode) and year, display the total number of rentals, total revenue, average revenue per rental, and average number of kilometers traveled per rental.
- For models of type 'economy' (Type attribute) and brand 'Toyota', separately by year and rental departure region, show the percentage of the number of rentals compared to the total number of rentals for each rental departure state and year.

Given the above logical schema, answer the following requests:

1. Define a materialized view with the CREATE MATERIALIZED VIEW, that can be used to efficiently answer all three of the above queries (i.e., a, b, c). Specifically, define the query in SQL associated with BLOCK A in the following instruction

```
CREATE MATERIALIZED VIEW ViewRent
BUILD IMMEDIATE
REFRESH FAST ON COMMIT
AS
 Block A
```

2. Assume that the management of the materialized view (derived table) is carried out by means of triggers. Write the trigger to propagate to the ViewRent materialized view the changes due to the insertion of a new record into the CAR-RENT fact table

DEFINITION OF INTERESTED QUERY

Query a

```
SELECT
    T.3-Months,
   SUM(SUM(CR.NumRents)) OVER(PARTITION BY T.Year ORDER BY T.3-Months ROWS UNBOUNDED
PRECEDING)
FROM
   CAR-RENT CR, CAR-MODEL CM, JUNK-RENT JR, TIME T
WHERE
   CR.IDModel=CM.IDModel AND
   CR.IDJR=JR.IDJR AND
   CR.IDTime=T.IDTime AND
   CM.Brand='Audi' AND
   JR.ClientType='business'
GROUP BY
   T.3-Months, T.Year
```

Query b

```
SELECT
   JR.PaymentMethod, T.Year,
   SUM(CR.NumRents),
   SUM(CR.Revenue),
   SUM(CR.Revenue)/SUM(CR.NumRents)
   SUM(CR.KMTraveled))/SUM(CR.NumRents)
FROM
   CAR-RENT  CR, CITY C, JUNK-RENT JR, TIME T
WHERE
   CR.IDDepartureCity=C.IDCity AND
   CR.IDJR=JR.IDJR AND
   CR.IDTime=T.IDTime AND
   C.Region='Piedmont'
GROUP BY
   JR.PaymentMethod, T.Year
```

Query c

```
SELECT
    T.Year, C.Region,
    100 * SUM(CR.NumRents)/SUM(SUM(CR.NumRents)) OVER(PARTITION BY C.Country, T.Year)
FROM
    CAR-RENT CR, CAR-MODEL CM, TIME T, CITY C
WHERE
    CR.IDModel=CM..IDModel AND CR.IDtime=T.IDTime AND CM:IDDepartureCity=C.IDCity
    CM.Type = 'economy' AND CM.Brand='Toyota'
GROUP BY
    T.Year, C.Region, C.Country
----
DEFINITION OF MW

SELECT
   JR.ClientType, JR.PaymentMethod,
   CM.Brand, CM.Type,
   T.3-Months, T.Year,
   CD.Region, CD.Country
   SUM(CR.NumRents) AS TotNumRents,
   SUM(CR.Revenue) AS TotRevenue,
   SUM(CR.KMTraveled) AS TotKMTraveled
FROM
    CAR-RENT CR, CAR-MODEL CM, JUNK-RENT JR, TIME T, CITY CD
WHERE
    CR.IDModel=CM.IDModel AND CR.IDJR=JR.IDJR AND CR.IDTime=T.IDTime AND
CR.IDDepartureCity=CD.IDCity
GROUP BY
    JR.ClientType, JR.PaymentMethod,
    CM.Brand, CM.Type,
    T.3-Months, T.Year,
    CD.Region, CD.Country

ID {ClientType, PaymentMethod, Brand, Type, T.3-Months, CD.Region }
----
DEFINTION OF TRIGGER

CREATE OR REPLACE TRIGGER ViewRentManagment
AFTER INSERT ON CAR-RENT
FOR EACH ROW
DECLARE
      My_clientType VARCHAR(30),
      My_PaymentMethod VARCHAR(30),
      My_brand VARCHAR(30),
      My_type VARCHAR(30),
```

```
     My_3M VARCHAR(7),
     My_1Y VARCHAR(4),
     My_region VARCHAR(30),
     My_country VARCHAR(30),
     N INTEGER
BEGIN

    SELECT JR.ClientType, JR.PaymentMethod INTO My_clientType, My_paymentMethod
    FROM JUNK-RENT JR
    WHERE JR.IDJR=:NEW.IDJR;

    SELECT CM.Brand, CM.Type INTO My_Brand, My_type
    FROM CAR-MODEL CM
    WHERE CM.IDModel=:NEW.IDModel;

    SELECT T.3-Months, T.Year INTO My_3M, My_1Y
    FROM TIME T
    WHERE T.IDTime=:NEW.IDTime;

    SELECT Region, Country INTO My_region, My_country
     FROM CITY CD
    WHERE CD.IDCity=:NEW.IDDepartureCity;

    SELECT
    FROM ViewRent VR
     WHERE VR.ClientType=My_ClientType AND VR.PaymentMethod=My_paymentMethod AND
VR.Brand=My_brand AND VR.Type=My_type AND VR.3-Months=My_3M AND VR.Region=My_region;

     IF ( N = 0 ) THEN
         INSERT INTO(...)
                        VALUES(My_ClientType,My_paymentMethod,My_brand, My_type, My_3M, My_1Y, My_region,
My_country, NEW.NumRents, .NEW.Revenue, :NEW.KmTraveled)
     ELSE
          UPDATE ViewRent VR
          SET VR.TotNumRents=VR.TotNumRents + :NEW.NumRents, VR.TotRevenue=VR.TotRevenue +
:NEW.Revenue, VR.TotKMTraveled=VR.TotKMTraveled + :NEW.KMTraveled
          WHERE VR.ClientType=My_ClientType AND VR.PaymentMethod=My_paymentMethod AND
VR.Brand=My_brand AND VR.Type=My_type AND VR.3-Months=My_3M AND VR.Region=My_region;
     END If

END
```

Commento:

DEFINTION OF TRIGGER

```
CREATE OR REPLACE TRIGGER ViewRentManagment
AFTER INSERT ON CAR-RENT
FOR EACH ROW
DECLARE
      My_clientType VARCHAR(30),
      My_PaymentMethod VARCHAR(30),
      My_brand VARCHAR(30),
      My_type VARCHAR(30),
      My_3M VARCHAR(7),
      My_1Y VARCHAR(4),
      My_region VARCHAR(30),
      My_country VARCHAR(30),
      N INTEGER
BEGIN

    SELECT JR.ClientType, JR.PaymentMethod INTO My_clientType, My_paymentMethod
    FROM JUNK-RENT JR
    WHERE JR.IDJR=:NEW.IDJR;

    SELECT CM.Brand, CM.Type INTO My_Brand, My_type
    FROM CAR-MODEL CM
    WHERE CM.IDModel=:NEW.IDModel;

    SELECT T.3-Months, T.Year INTO My_3M, My_1Y
    FROM TIME T
    WHERE T.IDTime=:NEW.IDTime;

    SELECT Region, Country INTO My_region, My_country
     FROM CITY CD
    WHERE CD.IDCity=:NEW.IDDepartureCity;

    SELECT COUNT(*) INTO N
    FROM ViewRent VR
     WHERE VR.ClientType=My_ClientType AND VR.PaymentMethod=My_paymentMethod AND
VR.Brand=My_brand AND VR.Type=My_type AND VR.3-Months=My_3M AND VR.Region=My_region;

     IF ( N = 0 ) THEN
         INSERT INTO(...)
                        VALUES(My_ClientType,My_paymentMethod,My_brand, My_type, My_3M, My_1Y, My_region,
My_country, NEW.NumRents, .NEW.Revenue, :NEW.KmTraveled)
     ELSE
          UPDATE ViewRent VR
          SET VR.TotNumRents=VR.TotNumRents + :NEW.NumRents, VR.TotRevenue=VR.TotRevenue +
:NEW.Revenue, VR.TotKMTraveled=VR.TotKMTraveled + :NEW.KMTraveled
```

WHERE VR.ClientType=My_ClientType AND VR.PaymentMethod=My_paymentMethod AND VR.Brand=My_brand AND VR.Type=My_type AND VR.3-Months=My_3M AND VR.Region=My_region;

END If

END

Domanda **12**

Risposta corretta

Punteggio ottenuto 1,50 su 1,50

1.5 points (-15% penalty for each wrong answer)

The single linkage policy states that the distance between two clusters X and Y is calculated as:

$dist(X,Y) = min(dist(x,y))$

where $x \in X, y \in Y$ and $dist(x,y)$ is the Manhattan distance between two points $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_n)$, defined as $dist(x,y) = \sum_{i=1}^{n} |x_i - y_i|$.

Given the following points in 3 dimensions:

A. 1,2,3
B. 2,4,4
C. 0,3,2
D. 4,9,4
E. 4,7,7

How are the points aggregated to get 2 clusters if we use the previous policy in agglomerative hierarchical clustering?

- a. {A, C, E, B} {D}
- b. {A, C, D} {B, E}
- c. {A} {D, E, C, B}
- d. No answer is correct
- e. {A, C, B} {D, E} ✔
- f. {A, C, E} {D, B}
- g. {A, B, C, D} {E}

Risposta corretta.

La risposta corretta è:
{A, C, B} {D, E}

Domanda **13**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Conceptual schema (1 point, -15% penalty for each wrong answer)

We want to analyze information about food drives organized during the year on the Italian territory by a charity. These events aim at collecting and subsequent redistribution of food items to charitable organizations in the territory.

Food drives are held several times a year, specifying the types of products to be collected, such as baby food and/or canned tuna and meat. To support the food drive, customers can purchase products at various stores. The purchased products are then handed over to volunteers outside the store. Customers who have purchased products can then respond to a short interview (anonymously and not compulsorily) for the purpose of collecting certain statistical information such as the customer's age, how the customer learned about the initiative (e.g., through social platforms and/or through the press), whether the customer is a member of the association, and the customer's occupation (e.g., whether the customer is a student, employee, or retiree).

In addition to the support received from people who buy the products, each food drive also benefits from the support of some organizations that donate funds to the food drive, such as banking foundations, companies, etc.

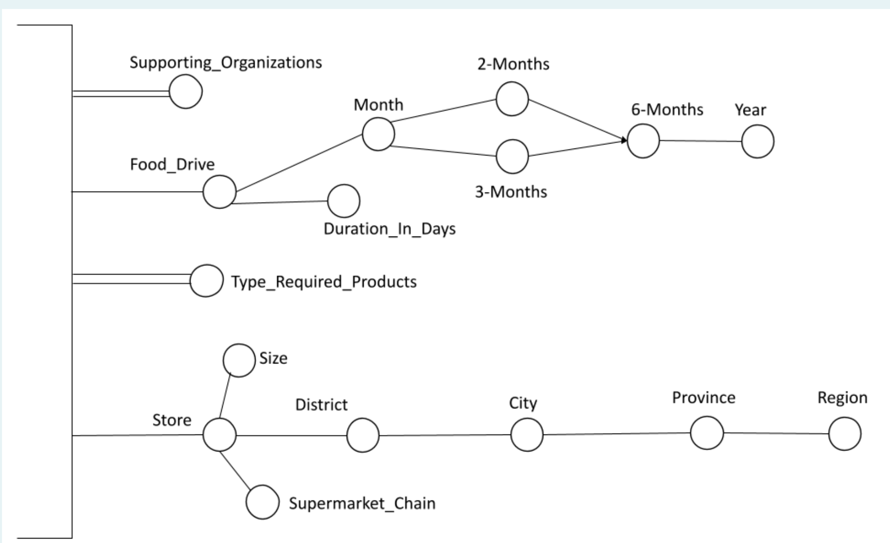
For each food drive, the month in which it is carried out is known, together with the indication of the corresponding 2-month period, 3-month period, 6-month period and year; the duration of the food drive in days is also known.

We want to analyze the average amount spent per customer and the average quantity purchased per customer based on the following information:

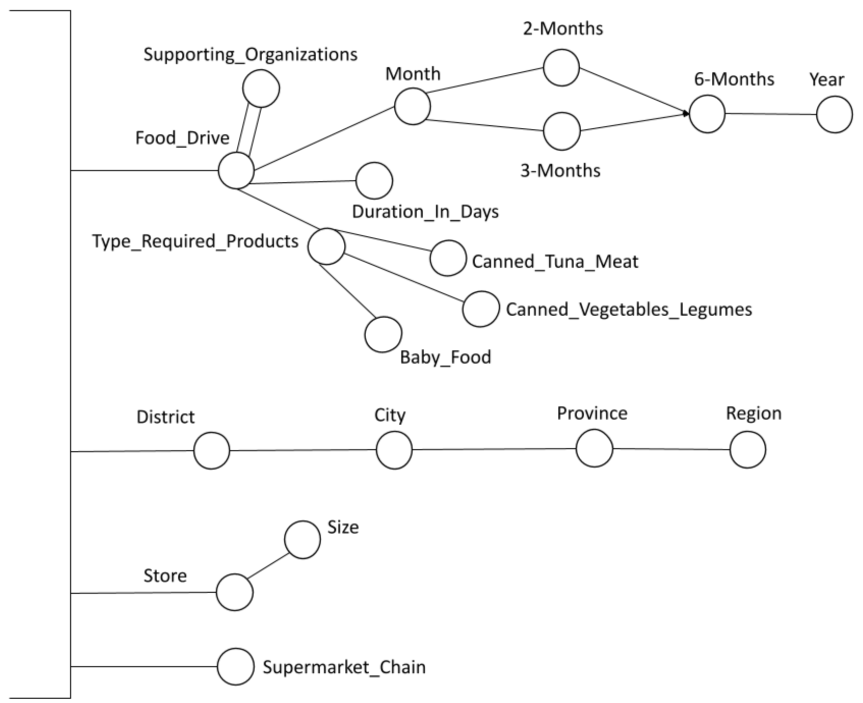
- food drives carried out. Each food drive is characterized by the type of products requested (one or more values among baby food, canned vegetables and legumes, canned tuna and meat) and by the list of organizations that supported the food drive (such as banking foundations, companies, etc.). For each food drive, the month in which it is carried out is known, together with the indication of the corresponding 2-month period, 3-month period, 6-month period, and year; the duration of the food drive in days is also known.
- stores where customers can buy products to support the food drive. For each store, the geographical location is known in terms of city district, city, province and region. The size of the store (a value between small, medium, large) and the supermarket chain to which it belongs are also known.
- type of product purchased by the customer (a value between baby food, canned vegetables and legumes, canned tuna and meat)
- indication whether the customer is a supporting member of the charity or not (a value between supporter and non-supporter)
- city, province, region and country of the customer who made the purchase
- indication of how the customer learned about the food drive (one or more values among social platforms, press, word of mouth, other)
- customer age range (a value between under 21, between 21 and 40, between 41 and 60, over 60) and customer occupation (a value between student, employee, retiree)
- date, month, year in which the customer made the purchase and the time slot (a value between morning, early afternoon, evening)

Select, from the dimensions proposed below, those that meet the requirements described in the problem specifications.

a.

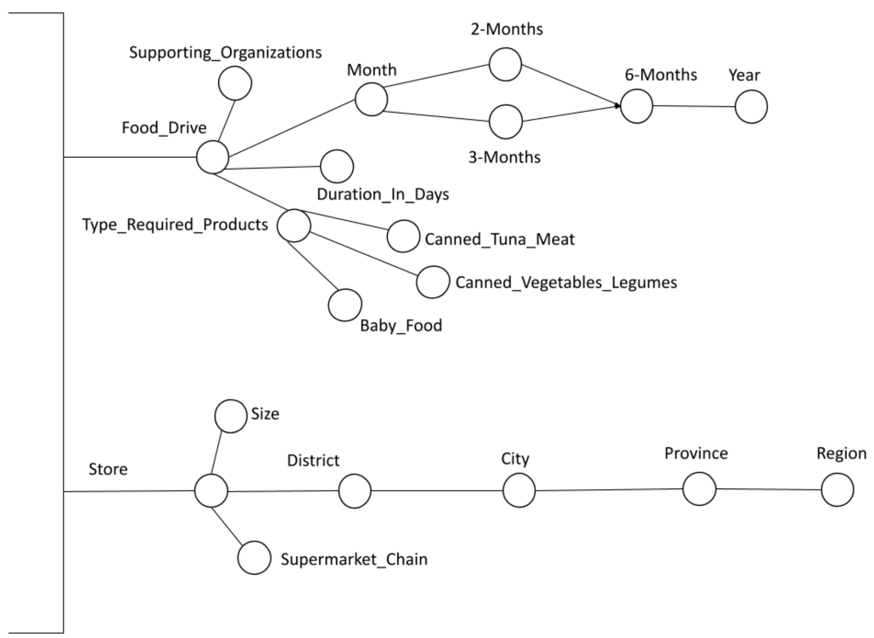


b.



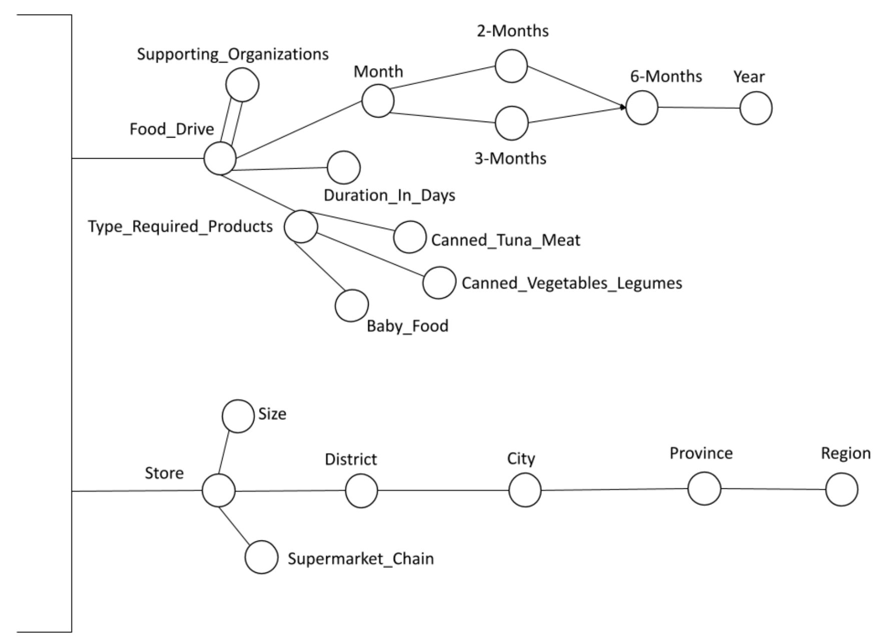
2-Months
Supporting_Organizations
Month
6-Months
Year
Food_Drive
3-Months
Duration_In_Days
Type_Required_Products
Canned_Tuna_Meat
Canned_Vegetables_Legumes
Baby_Food
District
City
Province
Region
Size
Store
Supermarket_Chain

c.

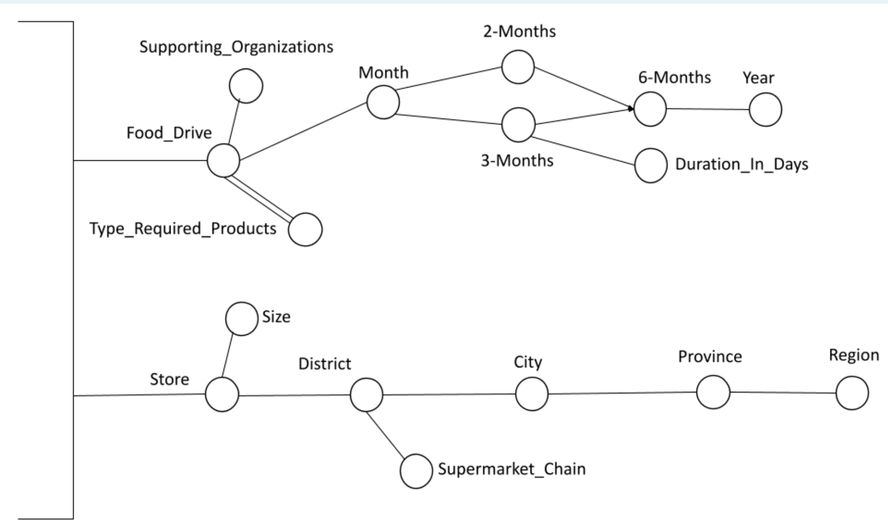


2-Months
Supporting_Organizations
Month
6-Months
Year
Food_Drive
3-Months
Duration_In_Days
Type_Required_Products
Canned_Tuna_Meat
Canned_Vegetables_Legumes
Baby_Food
Size
Store
District
City
Province
Region
Supermarket_Chain

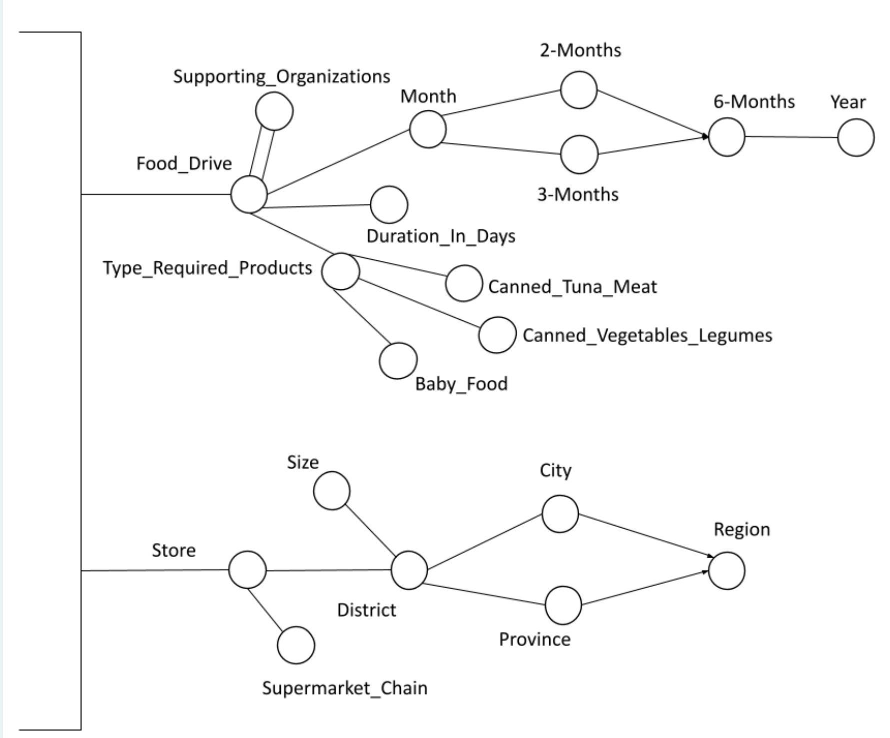
d.


Supporting_Organizations
2-Months
Month
6-Months
Year
Food_Drive
3-Months
Duration_In_Days
Type_Required_Products
Canned_Tuna_Meat
Canned_Vegetables_Legumes
Baby_Food
Size
District
City
Province
Region
Store
Supermarket_Chain

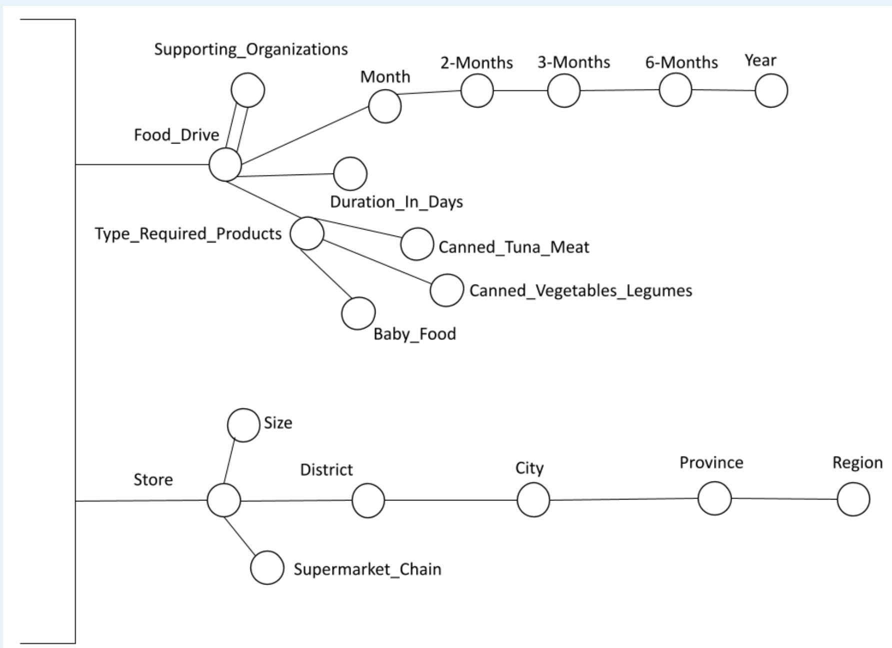
e.


2-Months
Supporting_Organizations
Month
6-Months
Year
Food_Drive
3-Months
Duration_In_Days
Type_Required_Products
Size
District
City
Province
Region
Store
Supermarket_Chain

f.

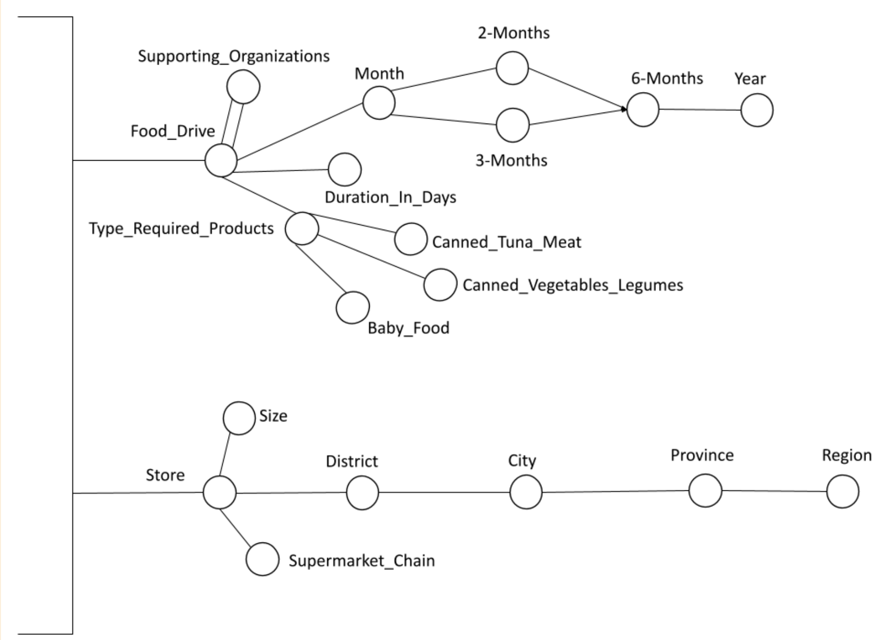


g.



Risposta corretta.

La risposta corretta è:



Domanda **14**

Risposta non data

Punteggio max.: 2,00

2 points (-15% penalty for each wrong answer)

The following transactions are given:

1. A, B, C, D
2. A, B, C, E
3. B, C, D
4. A, B, D, E, F
5. A, B, C, D, E, F
6. B, C, D, E, F
7. A, B, E
8. A, B, C, D, E, F
9. A, B, C, D
10. A, B, C, D, E, F

Construct an **FP-Tree** from the listed transactions, using a **MinSup >= 2**. An itemset is considered frequent if it appears in at least 2 transactions.

In case of items with equal support, lexicographic order is used (e.g., if items A and B have the same support, then A precedes B).

After constructing the FP-Tree, identify the patterns and their supports present in the **FE-CPB** (FE Conditional Pattern Base). Which of the following sets of **Pattern: Support** pairs is the FE-CPB? (Where pattern is a set of items).

- a. {DCAB: 3, DAB: 1}
- b. {BD: 5}
- c. {DCAB: 3, DAB: 1, DCB: 1, CAB: 1}
- d. {DCAB: 3, DAB: 1, CAB: 1}
- e. No answer is correct
- f. {DCAB: 3, DAB: 1, DCB: 1, AB: 1}

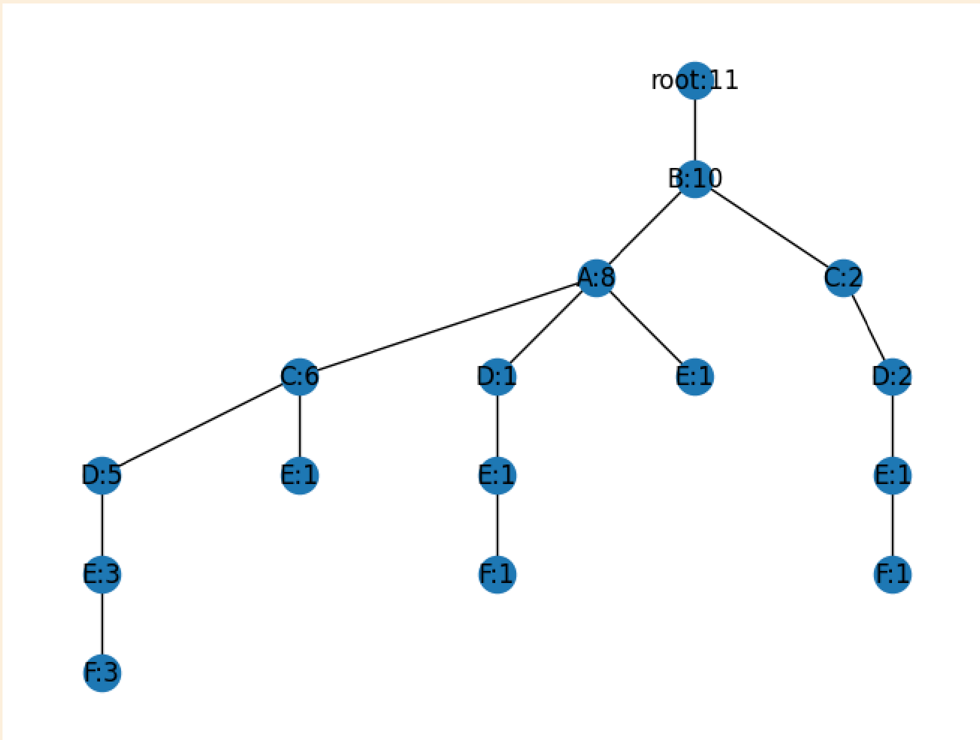
g. {DCAB: 3, DAB: 1, AB: 1}

Risposta errata.

F-CPB -> EDCAB: 3, EDAB: 1, EDCB: 1

F-CHT -> B: 5, D: 5, E: 5, A: 4, C: 4

FE-CPB -> BD: 5



La risposta corretta è:
{BD: 5}

Domanda 15
Risposta errata
Punteggio ottenuto -0,14 su 1,00

1 point (-15% penalty for each wrong answer)

Documentation

match: Filters documents based on a specified query predicate. Matched documents are passed to the next pipeline stage. Sintax: { $match: { <query predicate> } }
addFields: Adds new fields to documents. $addFields outputs documents that contain all existing fields from the input documents and newly added fields. Sintax: { $addFields: { <newField>: <expression>, ... } }
count: Counts and returns the total number of items in an array. Sintax: { $size: <expression> }

Text

A streaming service stores its data in a MongoDB collection called movies. Each document represents a movie, including genre, release year, cast, ratings, and viewer statistics. The structure of a document is as follows:

```
{
        "title": "The Adventure Begins",
        "genre": "Adventure",
        "releaseYear": 2020,
        "cast": ["John Actor", "Jane Star"],
        "ratings": [
                {"user": "Alice", "rating": 5},
                {"user": "Bob", "rating": 4}
        ],
        "viewers": [
                {"location": "USA", "count": 150000},
                {"location": "Canada", "count": 30000}
        ]
}
```

Which query can be used to show only the titles of movies released after 2015 in the genre “Adventure,” with an average rating of 4 or higher, and that had at least 200,000 total viewers overall?

a.

```
db.movies.aggregate([
  {
    $match: {
      genre: "Adventure",
      releaseYear: { $gt: 2015 }
    }
  },
  {
    $project: {
      title: 1,
      averageRating: { $avg: "$ratings.rating" },
      totalViewers: { $sum: "$viewers.count" }
    }
  }
]);
```

b.

```
db.movies.aggregate([
  {
    $addFields: {
      averageRating: { $avg: "$ratings.rating" },
      totalViewers: { $sum: "$viewers.count" }
    }
  },
  {
    $match: {
      genre: "Adventure",
      releaseYear: { $gt: 2015 },
      averageRating: { $gte: 4 },
      totalViewers: { $gte: 200000 }
    }
  },
  { $project: { title: 1, _id: 0 } }
]);
```

c.
```
db.movies.aggregate([
  {
    $match: {
      genre: "Adventure",
      releaseYear: { $gt: 2015 },
      averageRating: { $gte: 4 },
      totalViewers: { $gte: 200000 }
    }
  },
  { $project: { title: 1, _id: 0 } }
]);
```

d.
```
db.movies.aggregate([
  {
    $addFields: {
      averageRating: { $avg: "$ratings.rating" },
      totalViewers: { $sum: "$viewers.count" }
    }
  },
  {
    $match: {
      genre: "Adventure",
      releaseYear: { $gt: 2015 },
      averageRating: { $gte: 4 },
      totalViewers: { $gte: 200000 }
    }
  },
]);
```

e.
```
db.movies.aggregate([
  {
    $match: {
      genre: "Adventure",
      releaseYear: { $gt: 2015 }
    }
  },
  {
    $addFields: {
      averageRating: { $avg: "$ratings.rating" },
      totalViewers: { $sum: "$viewers" }
    }
  },
  {
    $match: {
      averageRating: { $gte: 4 },
      totalViewers: { $gte: 200000 }
    }
  },
  { $project: { title: 1, _id: 0 } }
]);
```
✖

f. No answer is correct

Risposta errata.

La risposta corretta è:

```
db.movies.aggregate([
  {
    $addFields: {
      averageRating: { $avg: "$ratings.rating" },
      totalViewers: { $sum: "$viewers.count" }
    }
  },
  {
    $match: {
      genre: "Adventure",
      releaseYear: { $gt: 2015 },
      averageRating: { $gte: 4 },
      totalViewers: { $gte: 200000 }
    }
  },
  { $project: { title: 1, _id: 0 } }
]);
```